

# Hands-on exercises 5: Ionization of stellar material

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And, once again, mostly analytical exercises, but with some actual calculations so keep python handy!

**Problem 1:** Convince yourself that, for stellar atmospheres, the mean molecular mass depends on the degree of ionization. For given gas pressure, assuming the gas consists only of hydrogen, calculate mass density for molecular, completely neutral and completely ionized hydrogen.

**Problem 2:** Using the results of the previous step - derive the equation of state (relationship between gas pressure, mass density and temperature) for pure hydrogen. Discuss what happens when we have more than one species.

**Problem 3:** Show that the adiabatic temperature gradient can be written as

$$\left. \frac{dT}{dr} \right|_a = -\frac{g}{c_p}, \quad (1)$$

where  $g$  is the acceleration due to gravity and  $c_p$  is the specific heat capacity at constant pressure.

## Useful physical constants

- $R_\odot = 696 \times 10^6 \text{ m}$
- $M_\odot = 1.989 \times 10^{30} \text{ kg}$
- $L_\odot = 3.83 \times 10^{26} \text{ W}$
- $T_\odot^{\text{eff}} = 5777 \text{ K}$
- $1 \text{ AU} = 1.496 \times 10^8 \text{ km}$
- $c = 2.997 \times 10^8 \text{ m/s}$
- $G = 6.674 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$
- $k = 1.38 \cdot 10^{-23} \text{ J/K}$
- $m_{\text{H}} = 1.67 \cdot 10^{-27} \text{ kg}$